

Reducing Demographic Bias in Chest X-Ray Diagnosis through Representation-Level Feature Filtering

Impact Statement

This work addresses fairness in medical imaging by examining demographic information encoded within CNN embeddings used for chest X-ray diagnosis. Unlike many existing approaches that focus mainly on retraining classifiers or modifying optimization strategies, this study investigates bias directly at the representation level. The proposed framework identifies embedding dimensions strongly associated with demographic attributes and removes them before disease classification. Experimental results show that this lightweight filtering strategy reduces demographic performance disparity while maintaining strong diagnostic accuracy. Because the method operates on pretrained embeddings, it can be integrated into existing medical AI pipelines without expensive retraining. The study highlights the importance of representation-level fairness analysis and provides a simple, computationally efficient direction for developing more reliable and equitable AI systems for healthcare.